



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch : III Semester B.E CSE B
Course Code & Title : CS3352 & Foundations of Data Science
Name of the Faculty member : Mrs. P.Sabeena Burvin, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit V / Histograms and Legends
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO 1,2,3,5,6,7,8,9,10,11
- **Activity Chosen:** Collaborative Learning
- **Date of Implementation:** 16.10.2025
- **Justification:**

Collaborative learning is an instructional approach in which learners work together in pairs or small groups to achieve shared academic goals. Implementing this approach is justified on pedagogical, cognitive, and social grounds.

- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
 - The collaborative learning activity provide students the best opportunities to practice the tasks in an authentic way, including interactions and working with other people.
 - The activity was conducted for the duration of 20 minutes.
 - Initially the teams are formed based on self-selection in which the students are form their own groups of size 2.
 - Once the teams are formed, they are discussed the various key distribution and key management methods.
 - The team members were discussed with them. It is shown in Fig.1.
 - The instructor monitored all the teams and provided any clarification if needed.
 - After 20 minutes of discussion the faculty member called every team to present their Knowledge. It shown in Fig 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO12	PSO1	PSO3
CO1	2	2	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO12	PSO1	PSO3
	2	2	3	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of visualization libraries in python to interpret data for solving problems	Identify the need of visualization on libraries in python to interpret and explore data	Able to design solutions for any applications using visualization on libraries in python	Able to analyze and apply visualization on libraries in python and provide valid conclusions	Ability to recognize the need for visualization on libraries in python in real world application	Exhibit knowledge in visualization libraries for data science application development

- **Images / Screenshot of the practice:**



Figure 1: Team Discussion



Figure 2: Visualization presented by Parvathi Devi

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students reported improved confidence in explaining concepts.
- Collaborative planning strengthened teamwork and problem-solving.
- They felt that through such learning, they can explore more.

- ❖ ***Benefit of the practice:***

- More collaboration time for students.
- Lessons became more relatable due to student-generated examples.
- Learners felt more ownership of knowledge.

- ❖ ***Challenges faced in implementation:***

- Some groups initially struggled with simplifying technical concepts.
- Time management during student-led sessions required guidance.
- A few students were hesitant to take the role of teacher, requiring encouragement.

References:

- ❖ <https://www.hunschool.org/resources/problem-based-learning>
- ❖ <https://www.monash.edu/learning-teaching/TeachHQ/Teachingpractices/collaborative-learning/how-to/collaborative-learning-processes>
- ❖ <https://tilt.colostate.edu/TipsAndGuides/Tip/180>
- ❖ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3748308/>

